closed on itself like a rubber band or it extends on and on forever), and its space warpage causes any circle around it to have a circumference divided by diameter slightly less than π.

critical circumference. The circumference of the horizon of a black hole; the circumference inside which an object must shrink in order for it to form a black hole around itself. The value of the critical circumference is 18.5 kilometers times the mass of the hole or object in units of the mass of the Sun.

curvature of space or spacetime. The property of space or spacetime that makes it violate Euclid's or Minkowski's notions of geometry; that is, the property that enables straight lines that are initially parallel to cross.

Cyg A. Cygnus A; a radio galaxy that looks like (but is not) two colliding galaxies. The first radio galaxy to be firmly identified.

Cyg X-1. Cygnus X-1; a massive object in our galaxy that is probably a black hole. Hot gas falling toward the object emits X-rays observed on Earth.

dark star. A phrase used in the late eighteenth and early nineteenth centuries to describe what we now call a black hole.

degeneracy pressure. Pressure inside high-density matter, produced by erratic, high-speed, wave/particle-duality-induced motions of electrons or neutrons. This type of pressure remains strong when matter is cooled to absolute zero temperature.

deuterium nuclei, or deuterons. Atomic nuclei made from a single proton and a single neutron held together by the nuclear force. Also called "heavy hydrogen" because atoms of deuterium have almost the same chemical properties as hydrogen.

differential equation. An equation that combines in a single formula various functions and their rates of change; that is, the functions and their "derivatives." By "solve a differential equation" is meant "compute the functions themselves from the differential equation."

Doppler shift. The shift of a wave to a higher frequency (shorter wavelength, higher energy) when its source is moving toward a receiver, and to a lower frequency (longer wavelength, lower energy) when the source is moving away from the receiver.

electric charge. The property of a particle or matter by which it produces and feels electric forces.